

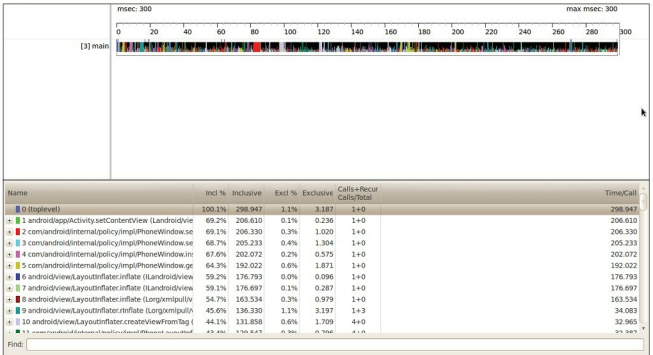


Figure 11: Traceview information [traceview.png]

- On the VM Heap tab, if you see, “No heap updates are available for this client,” then click the **Show Heap Updates** button at the top of the left pane, and then click the **Cause GC** button on the VM Heap tab.

You will now be able to see the heap update status, showing available memory on the device, along with some extra information about the heap.

Emulator

This tool is used to launch and control an emulator instance (an instance of an AVD created by the *android* utility). To launch an AVD, run...

```
emulator -avd <avd name>
```

For example, `emulator -avd myandy_2.2`. We can use the *emulator* utility to control various hardware simulations of the emulator instance, like networking, media, etc, using start-up options—additional parameters on the emulator command-line. The general syntax is `emulator -avd <avd name> <options> <values>`. Some of the commonly-used options are `-logcat`, to view the console log while Android is booting; `-shell` to open a root shell on the emulator instance; and `-no-boot-anim` to disable the Android logo animation during boot.

Networking in the emulator

When an emulator instance is powered up, it always runs behind a virtual router/firewall and hence is isolated from the development environment. IP addresses used by emulator

instances are in the range of 10.0.2.0 – 10.0.2.99.



Note: To access the development machine's localhost, it is by default set to 10.0.2.2.

Once the emulator instance is running, we can connect to the instance's control console using a simple *telnet* connection like `telnet localhost <console_port>`. For example, `telnet localhost 5554`. Type *help* at the Android console for a list of available commands, as shown in Figure 10. Similarly, there are plenty of options available for the emulator; read about them at <http://goo.gl/Mmqh>.

Mksdcard

This tool lets you create an SD card disk image, with a FAT32 file system, which you can load into the emulator. The syntax is...

```
mksdcard -l <label> <size>[K|M] <file>
```

For example: `mksdcard -l S 512M saket.img`. To ‘insert’ this SD card image into an emulator instance, we use: `emulator @myandy_2.2 -sdcard saket.img`.

SQLite3

This command is used to enter the SQLite database system. You can use it to create and modify databases present in your emulator instance or device. To run the SQLite client, obtain a shell in the emulator instance and run the *sqlite3* command. For example: